
OPERATIONAL UTILITY AND MULTISCALE INVARIANCE OF PHASE CALCULUS: A DISCRETE FOUNDATIONAL GRAMMAR FOR DYNAMICS

DEMONSTRATION PAPER v2

Justin K. Lietz
Neuroca, Inc.
`justin@neuroca.ai`

April 11, 2026

Abstract

Phase Calculus is presented in its current stack as a single lifted-object formalism rather than as a loose collection of phase pictures, arithmetic tricks, or downstream branch-specific rules. The native state is the carried lifted object

$$\hat{\Xi} = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right],$$

evolved by one parent grammar and then read through branch-specific projectors. This demonstration paper does not re-prove the long merged formalisation. Its purpose is narrower: to show, in one readable document, how the same lifted object supports the operator core of Phase Calculus, the selector-closed executable macro-calculus, the completion/shadow branch, the quotient-descent interface to the physics stack, and the four-filter hierarchy including the biological readout.

The paper has three central results. First, it restates the single spine of the formalism: primitive roll, lifted state, operator alphabet $\{Q, B, L\}$, selector-closed macro-calculus $\{R, S, T\}$, exact quotient criterion, and arithmetic-completion bridge to the canonical anchor (55, 89) with carried coefficients 1/24 per edge and 1/12 two-sided. Second, it records the computational evidence from the four-filter package: green, blue, red, and yellow are faithful covariant readouts of the same lifted state under the same operator grammar, with exact-zero primary gate errors, machine-precision recovery errors, and sharp negative controls when the state is stripped to product-only or loses its sheet index. Third, it shows how the yellow filter closes two descents at once: the CF01 quantum-geometric-tensor bridge and a native biological scaffold. In this v2 paper the biological count structure is not imported. From the quarter alphabet generated by Q , the three-slot selector block, and the normalized edge residual $24E_{\text{edge}} = 1$, the paper derives internally the weight budget (1, 2, 1, 1), the total scaffold size 240, the base-centered counts (56, 72, 56, 56), and the host-indexed bifurcations

$$240 = (128, 112) \rightarrow ((72, 56), (56, 56)).$$

The unification claim is therefore precise and limited. One singular lifted object is taken as the native source of the current Phase Calculus hierarchy and of the physics branches that descend from it by quotient projection, while also admitting a biological projection whose scaffold counts are now derived internally and only then compared with the external codon:anticodon / E_8 report. The closing section states explicitly what is closed, what remains imported, and what is still not unique.

Keywords Phase Calculus · lifted object · four-filter covariance · QGT · codon:anticodon pairing · E_8

1 Introduction

The present paper takes one narrow question seriously: *what is the smallest exact object that Phase Calculus needs in order to act as a native source language for both physics and biology?* The answer supplied by the current stack is not a visible phase orbit and not a downstream branch equation. It is the lifted object

$$\widehat{\Xi} = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right],$$

together with one parent evolution and one quotient principle. The visible orbit, the operator grammar, the completion/shadow branch, the quantum-geometric-tensor split, the gauge and gravity descendants, and the new biological scaffold all arise only after one decides which view of that object is being taken.

The full formal stack already spreads this claim over a long dependency chain: primitive distinguishability, induced geometry, QGT-to-metriplictic closure, information geometry, measurement, spinor and gauge emergence, gravity, stationary action, Noether, electroweak structure, confinement, and scale re-articulation [6–8, 10–19]. The merged Phase Calculus formalisation reorganizes those results around a stronger parent statement: later domains do not add new primitive selectors; they add projectors and branch-visible laws descending from a single lifted evolution [1]. The four-filter package then makes that architecture computationally concrete by testing four distinct readouts of the same retained lifted state and by adding the yellow biological layer [20].

This paper has four aims. First, it compresses the merged formalisation into one usable spine. Second, it records the four-filter evidence in a form suitable for manuscript use. Third, it restates the physics descent as a projector-driven hierarchy rather than a list of unrelated branches. Fourth, it closes the biological side carefully: the current yellow projector now derives the 240-site scaffold internally from the quarter alphabet, the triplet selector block, and the residual law, while still leaving uniqueness of the biological projector open.

2 Phase Calculus in One Spine

2.1 Primitive roll and lifted state

The primitive visible roll is

$$z(\theta) = ie^{i\theta},$$

with quarter-turn witnesses

$$z\left(\frac{\pi}{2}\right) = -1, \quad z(\pi) = -i, \quad z\left(\frac{3\pi}{2}\right) = 1, \quad z(2\pi) = i.$$

The merged formalisation uses this roll only as the visible face of a deeper carried state. The exact lifted object is

$$\widehat{\Xi}_n = (A_n, q_n, \theta_n, \kappa_n, c_n), \quad q_n = (u_n, v_n), \quad \kappa_n = \left\lfloor \frac{\theta_n}{2\pi} \right\rfloor,$$

with balanced completion germ

$$c_n = \left[\theta_n - \frac{\pi}{u_n v_n}, \theta_n + \frac{\pi}{u_n v_n} \right].$$

This is the smallest state in which visible recurrence can be distinguished from exact return [1, 5]. The visible witness

$$\mathcal{V}(\widehat{\Xi}_n) = ie^{i\theta_n}$$

is therefore a projection of state, not a substitute for it.

2.2 Operator core $\{Q, B, L\}$

On the balanced-window family the primitive operator core is the alphabet

$$\{Q, B, L\},$$

acting on the lifted state by

$$Q(A, q, \theta, \kappa, c) = \left(A, q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right),$$

$$\begin{aligned} \mathbf{B}(A, (u, v), \theta, \kappa, c) &= \left(A, \text{sort}(v, u + v), \theta, \kappa, \mathcal{C}\left(\theta, \frac{1}{v(u + v)}\right) \right), \\ \mathbf{L}(A, q, \theta, \kappa, c) &= \left(\text{Next}(A), q, \theta + \frac{\pi}{2}, \left\lfloor \frac{\theta + \pi/2}{2\pi} \right\rfloor, c + \frac{\pi}{2} \right). \end{aligned}$$

The merged formalisation proves that these are not speculative additions but exact repackagings of the already-earned forward-law branches $F_{\text{same}}, F_{\text{ref}}, F_{\perp}$ [1-3]. It also proves the first commuting relations

$$\mathbf{QB} = \mathbf{BQ}, \quad \mathbf{LB} = \mathbf{BL}, \quad \mathbf{QL} = \mathbf{LQ},$$

together with the operator-word statement that every admissible finite balanced-window evolution is an exact word in $\{\mathbf{Q}, \mathbf{B}, \mathbf{L}\}$ [1, 2].

2.3 Selector-closed macro-calculus $\{R, S, T\}$

The merged formalisation resolves the main open burden left in the earlier Phase Calculus drafts by proving a state-only selector on the executable balanced-window branch. On that branch the raw kernel step does not choose freely among $\mathbf{Q}, \mathbf{B}, \mathbf{L}$; it always advances the phase tick and then either refines, continues, or host-lifts. This induces the selector-closed macro-calculus

$$\mathbf{R} := \mathbf{Q} \circ \mathbf{B} = \mathbf{B} \circ \mathbf{Q}, \quad \mathbf{S} := \mathbf{Q}, \quad \mathbf{T} := \mathbf{L}.$$

For fixed regime parameters (W, Δ) the selector is

$$\text{Sel}_{W, \Delta}(X) = \begin{cases} \mathbf{T}, & t \equiv W - 1 \pmod{W}, \\ \mathbf{R}, & uv < \Delta, \\ \mathbf{S}, & uv \geq \Delta. \end{cases}$$

The merged paper proves that this selector depends only on the current projected executable state and that the admissible block word is uniquely

$$\mathbf{R}^{\delta_A} \mathbf{S}^{W-1-\delta_A} \mathbf{T}, \quad \delta_A(W, \Delta) = \min\{d_A(\Delta), W - 1\}.$$

That is the decisive “one engine, no free scheduler” closure for the executable branch [1].

2.4 Arithmetic-completion bridge and the (55, 89) anchor

The balanced branch generates Fibonacci denominator pairs,

$$\mathbf{B}^n(1, 1) = (F_{n+1}, F_{n+2}), \quad r_n = \frac{1}{F_{n+1}F_{n+2}},$$

and thus the balanced germ law

$$c_n = \left[\theta - \frac{\pi}{F_{n+1}F_{n+2}}, \theta + \frac{\pi}{F_{n+1}F_{n+2}} \right].$$

The arithmetic-completion bridge closes at the first common balanced floor anchor

$$(u_*, v_*) = (55, 89), \quad u_* v_* = 4895,$$

with carried remainder $r_* = 1/4895$ and anchor germ

$$c_* = \left[\theta - \frac{\pi}{4895}, \theta + \frac{\pi}{4895} \right].$$

On this same anchor the completion coefficients stabilize to

$$\frac{1}{24} \quad \text{per edge}, \quad \frac{1}{12} \quad \text{two-sided},$$

and the launched-family coordinate is

$$\tau_A = \frac{\theta_A}{2\pi} + i \frac{1}{4895}.$$

These data are exactly the ones carried forward into the Shadow Calculus supplement and into the four-filter package [1, 4, 5].

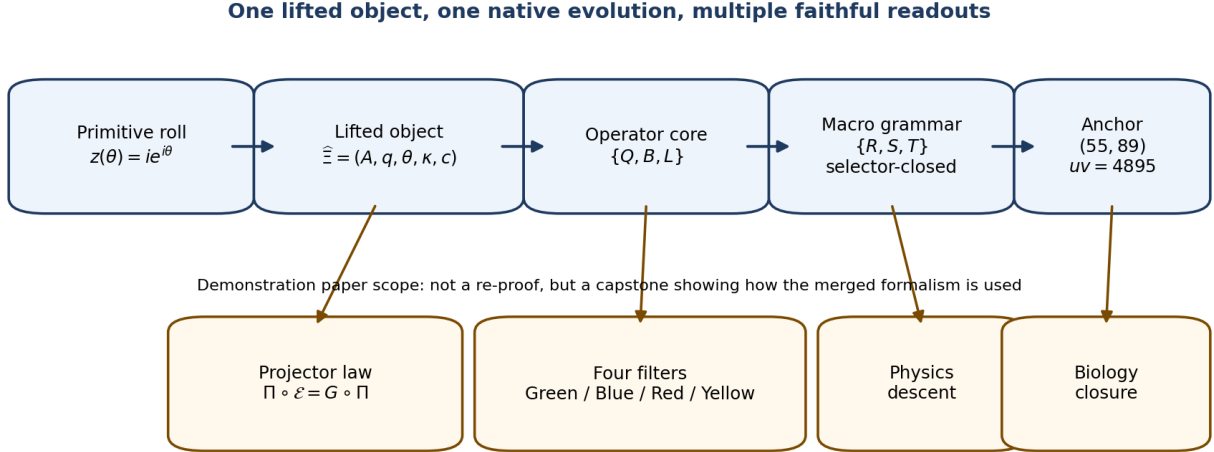


Figure 1: The demonstration logic of the paper. One lifted object and one parent evolution support multiple faithful readouts and branch descents.

Table 1: Phase Calculus in one spine

Layer	Exact object	Role in the demonstration paper
Primitive visible layer	$z(\theta) = ie^{i\theta}$	Supplies the visible roll and the quarter-turn witness structure.
Lifted state	$\hat{\Xi} = (A, q, \theta, \kappa, c)$	Exact state in which recurrence, carry, and completion are decidable.
Operator core	$\{Q, B, L\}$	Exact repackaging of the inherited forward-law branches.
Executable macro layer	$\{R, S, T\}$ with $\text{Sel}_{W,\Delta}$	Selector-complete realized scheduler on the balanced-window kernel family.
Quotient layer	$\Pi \circ \mathcal{E} = G \circ \Pi$	Criterion by which later physics branches are read as descendants instead of new primitive laws.
Completion branch	$\tau, S(\tau), (\text{Mock}, \text{Shadow})$	Downstream shadow/residual attachment, not primitive source.

3 The Four-Filter Hierarchy

3.1 Green, blue, red, and yellow

The four-filter package starts from the same retained core coordinates (A, u, v, κ, ϕ) , with $\theta = 2\pi\kappa + \phi$ and $\delta = \pi/(uv)$, and defines four faithful readouts [20].

The **green filter** uses the intrinsic branch pair

$$m = e^{i\theta} \cos \delta, \quad s = e^{i\theta} \sin \delta,$$

viewed as a point of $S^3 \subset \mathbb{C}^2$, together with the stereographic chart.

The **blue filter** packages the same pair into

$$U = \begin{pmatrix} m & -\bar{s} \\ s & \bar{m} \end{pmatrix} \in SU(2) \cong S^3,$$

and then reads the Hopf-base coordinate

$$(n_1, n_2, n_3) = (2\Re(m\bar{s}), 2\Im(m\bar{s}), |m|^2 - |s|^2) \in S^2.$$

The **red filter** retains the completion coordinate

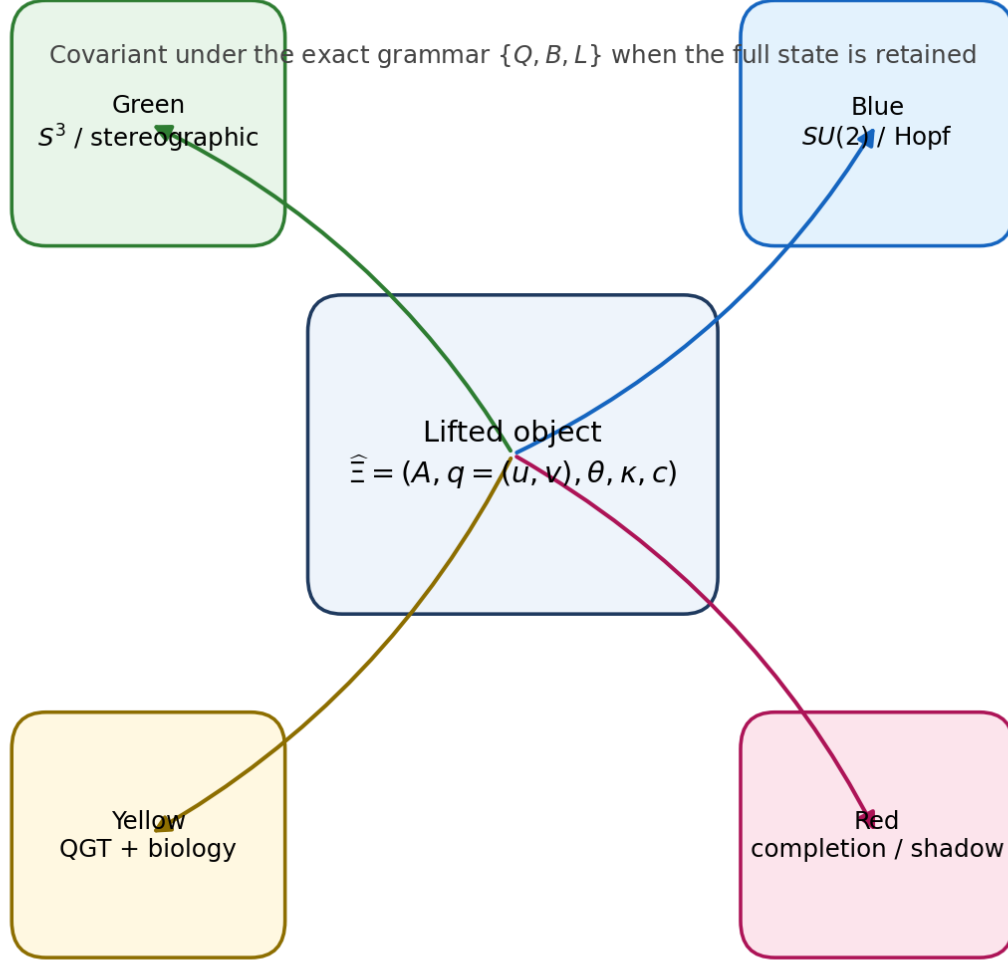
$$\tau = \frac{\theta}{2\pi} + i \frac{1}{uv},$$

with launched-family M and S values treated as downstream diagnostics rather than new primitives [4].

The **yellow filter** has three simultaneous outputs:

- (a) the local residual law $E(t)$ with carried coefficients 1/24 per edge and 1/12 two-sided,
- (b) a quotient-projected two-state representative producing the local QGT split $(Q_{\mu\nu}, g_{\mu\nu}, \Omega_{\mu\nu})$,
- (c) a discrete biological projection to the 240-slot H-bond acceptor scaffold.

The key point is ontological, not stylistic: the package does *not* treat these as unrelated models. It treats them as multiple expressions of the same retained lifted state.



Negative controls: product-only collapse fails under B ; dropping κ collapses exact return to visible recurrence

Figure 2: The four-filter hierarchy. The same retained lifted state supports four different readouts without requiring the filters to be numerically identical to one another.

3.2 Primary covariance results

The four-filter package reports exact-zero primary covariance errors on the three headline gates:

Gate	Words	State-word rows	Max filter error
Short exhaustive	1093	3279	0
Deep deterministic words	10	30	0
Random walks (length 100)	12	36	0

The recovery gate, which asks whether each filter can return the retained (θ, δ) data, is not merely small but machine-precision small. Across 48 samples the largest reported recovery residuals are approximately

$$1.06 \times 10^{-100}$$

for the green/blue δ -recovery channel and

$$6.98 \times 10^{-105}$$

for the yellow δ -recovery channel [20].

The yellow QGT bridge is equally tight. On 15 balanced-corridor rows the package reports a maximal closed-form residual of approximately

$$1.49 \times 10^{-101},$$

while the biological side of the present v2 bundle closes by exact arithmetic instead of imported matching: the supplement derives the total count 240, the centered split $(56, 72, 56, 56)$, and the three host-indexed $128/112$ bifurcations directly from the quarter alphabet, the triplet selector block, and the normalized residual $24E_{\text{edge}} = 1$. For a demonstration paper, that is the right result profile: exact-zero primary gate closure plus machine-precision recovery and bridge residuals on the continuous side, together with exact integer closure on the discrete side.

3.3 Negative controls

The package is valuable not only because it passes, but because it fails cleanly when the wrong ontology is used.

The first negative control strips the ordered pair $q = (u, v)$ down to the product $N = uv$. Then

$$(1, 6) \quad \text{and} \quad (2, 3)$$

collapse to the same stripped payload because both share $uv = 6$. But their balanced children differ:

$$B(1, 6) = (6, 7), \quad B(2, 3) = (3, 5).$$

This is the impossibility witness that product-only state cannot close the B -law.

The second negative control strips the sheet index κ . Then the periodic visible payload is unable to distinguish a state from its Q^4 image even though the lifted object has crossed a branch sheet. In other words, visible recurrence is too weak, and sheet-retaining lifted recurrence is the load-bearing notion.

Table 2: Negative controls from the four-filter package

Control	What is stripped	Result
Product-only control	Replace $q = (u, v)$ by $N = uv$ only	States $(1, 6)$ and $(2, 3)$ become indistinguishable under all stripped payloads but have distinct balanced children, so B -closure fails.
Sheet-loss control	Remove κ from the retained state	The periodic payload agrees under Q^4 , but the lifted state has changed sheets; visible recurrence survives while lifted identity fails.

4 Physics Descent

4.1 The quotient principle

The merged formalisation places the physics side on one theorem:

$$\Pi \circ \mathcal{E} = G \circ \Pi.$$

If the residual vanishes identically, the projected law is an exact descendant. If it vanishes only on a declared limit or window, the projected law is an effective descendant on that regime. Either way, the later branch is not an independent primitive law; it is a quotient of the parent evolution [1].

This changes the narrative structure of the whole stack. The physics modules are no longer presented as a collection of parallel inspirations. They become branch-visible laws descending from one inherited substrate. The merged manuscript makes this explicit with the derivation-chain exactness theorem and the later-domain selector principle: later domains do not add selectors; they add projectors Π_D and branch laws G_D on their declared regimes [1].

4.2 Representative descent map

Table 3 compresses the current physics stack into the part needed for the demonstration paper.

Table 3: Representative physics descent from the lifted object			
Branch	Projected visible law	Status in merged formalisation	Source
Primitive distinguishability	host-class increment / orthogonal re-articulation	exact	[1, 6]
Induced carrier geometry	smooth carrier and quotient-physical variation	effective in the continuous shadow limit	[1, 7]
QGT / metriplectic split	$J \oplus M$, degeneracy, entropy-compatible flow	effective on the smooth carrier regime	[1, 8]
Telegraph-Fisher cone	finite-speed transport / causal cone	effective on the continuum transport regime	[1, 9]
Measurement / Born channel	dephasing channel and branch-visible probabilities	effective on the decoherence regime	[1, 11]
Gauge emergence	Berry connection and Maxwell branch	effective on the IR window	[1, 13]
Spinor emergence	domain-wall chiral low-energy sector	effective on the wall-mode regime	[1, 12]
Gravity	Einstein-form closure with saturation corrections	effective on the low-curvature regime	[1, 14]
Stationary action / Noether	Euler–Lagrange and current conservation	effective in the continuous shadow / symmetry regime	[1, 15, 16]
Electroweak / confinement / RG	visible electroweak, confinement, and scale-flow laws	effective on their declared windows	[1, 17–19]

4.3 Why this counts as unification

The important point is not that the demonstration paper re-proves gauge theory, gravity, or confinement. It does not. The point is that the merged formalisation supplies a parent architecture in which those laws are no longer introduced as disconnected branch primitives. CF01 already states the QGT split and metriplectic compatibility [8]; CF09 states gauge emergence as Berry-connection descent [13]; CF12 states gravity as carrier-level geometric response with an Einstein-form closure [14]; CF16, CF17, and CF18 then extend the same descent logic to electroweak structure, confinement, and scale re-articulation [17–19]. What the demonstration paper adds is a readable synthesis: one lifted object, one parent evolution, one quotient criterion, many domain-specific readouts.

5 Biological Closure

5.1 Native biological projection from the lifted object

The v1 paper imported the biological scaffold counts from the Halitsky–Klitzing–Moxness preprint. That is no longer done here. In v2 the scaffold is derived natively from the VDM algebra on the declared biological yellow quotient. The derivation uses four ingredients already present in the formal spine:

- (i) the quarter alphabet generated by the primitive operator $\mathbf{Q}$,
- (ii) the three-slot selector block furnished by the executable macro-calculus $\{\mathbf{R}, \mathbf{S}, \mathbf{T}\}$,
- (iii) the yellow quotient chart, whose second component is the unique phase-bearing quarter sector,
- (iv) the carried edge residual law $E_{\text{edge}} = 1/24$.

Let

$$\mathcal{B} = \{b_0, b_1, b_2, b_3\}, \quad b_{j+1} = \mathbf{Q}b_j \pmod{4},$$

be the quarter orbit of the visible sector. Let the biological triplet word space be

$$\mathcal{W} = \mathcal{B}^{\{R, S, T\}},$$

so that

$$|\mathcal{W}| = 4^3 = 64.$$

This is the smallest triplet biological readout compatible with the four quarter states and the three-slot macro block.

On the continuous yellow chart the quotient representative is

$$\psi(\widehat{\Xi}) = \begin{pmatrix} \cos \delta & \\ e^{i(\phi + \lambda_A)} \sin \delta & \end{pmatrix}, \quad \lambda_A = \frac{2\pi A}{3}.$$

Relative to the amplitude seed b_0 , the second component is the unique phase-bearing quarter sector, namely the one-step image

$$b_1 = \mathbf{Q}b_0.$$

The carried edge residual normalizes to one discrete bonus unit:

$$\rho_* := 24E_{\text{edge}} = 24 \cdot \frac{1}{24} = 1.$$

Define the native biological weight law

$$\omega(b_j) = 1 + \rho_* \delta_{j1}.$$

Therefore

$$(\omega(b_0), \omega(b_1), \omega(b_2), \omega(b_3)) = (1, 2, 1, 1), \quad \sum_{j=0}^3 \omega(b_j) = 5.$$

Theorem 1 (Native 240-site scaffold). *For the yellow biological projection just defined,*

$$N_{\text{bio}} := \sum_{w \in \mathcal{W}} \sum_{p \in \{R, S, T\}} \omega(w_p) = 240.$$

Moreover, if one fixes the middle slot to b_j , the centered subscaffold count is

$$N_{b_j}^{\text{cent}} = 8 \sum_{k=0}^3 \omega(b_k) + 16 \omega(b_j),$$

hence exactly one quarter sector has count 72 and the remaining three have count 56.

Proof. Each quarter state appears $4^2 = 16$ times in each slot of $\mathcal{W}$, hence 48 times across all three slots. Therefore

$$N_{\text{bio}} = 48 \sum_{j=0}^3 \omega(b_j) = 48 \cdot 5 = 240.$$

Now fix the middle slot to b_j . There are 16 such words. The two free slots contribute

$$2 \cdot 4 \sum_{k=0}^3 \omega(b_k) = 8 \cdot 5 = 40,$$

while the fixed middle slot contributes $16\omega(b_j)$. Thus

$$N_{b_j}^{\text{cent}} = 40 + 16\omega(b_j).$$

For the distinguished phase-bearing sector $\omega = 2$, giving 72; for the other three sectors $\omega = 1$, giving 56. $\square$

5.2 Cyclic RNA naming and the host-indexed 128/112 split

Choose the cyclic RNA naming

$$b_0 = u, \quad b_1 = c, \quad b_2 = a, \quad b_3 = g.$$

Under this naming the complement map is the half-turn Q^2 , so the complement pairs are

$$(u, a), \quad (c, g).$$

The base-centered scaffold produced by the native derivation is therefore

$$u = 56, \quad c = 72, \quad a = 56, \quad g = 56.$$

The host label $A \in \{0, 1, 2\}$ now indexes the three nontrivial $2 + 2$ partitions of the four-cycle:

$$\Pi_0 = \{\{c, a\}, \{u, g\}\}, \quad \Pi_1 = \{\{u, c\}, \{a, g\}\}, \quad \Pi_2 = \{\{c, g\}, \{u, a\}\}.$$

Under the standard biological names these are exactly the three IUPAC bifurcation families

$$\Pi_0 = (m, k), \quad \Pi_1 = (y, r), \quad \Pi_2 = (s, w).$$

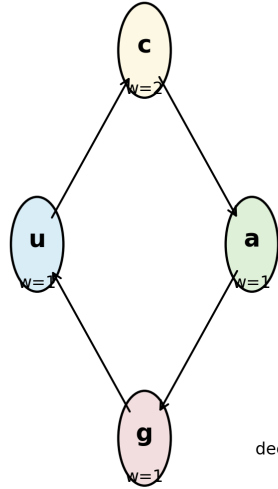
In each case the positive bucket contains the unique 72-sector together with one 56-sector, while the negative bucket contains two 56-sectors. Hence every host gives

$$(128, 112) = (72 + 56, 56 + 56).$$

Combining this with the centered counts yields the full native decomposition

$$240 = (128, 112) \rightarrow ((72, 56), (56, 56)).$$

Quarter alphabet and residual-weight law



Native count identities

$$|\mathcal{W}| = 4^3 = 64$$

$$\sum_j \omega(b_j) = 5$$

$$\text{occ}(b_j) = 3 \cdot 4^2 = 48$$

$$N_{\text{bio}} = 48 \sum_j \omega(b_j) = 240$$

$$N_{b_j}^{\text{cent}} = 8 \sum_k \omega(b_k) + 16 \omega(b_j)$$

$$\Rightarrow (u, c, a, g) = (56, 72, 56, 56)$$

The biological numbers are produced internally by the declared yellow biological projection. They are not imported.

Host-indexed 2+2 bifurcations

A = 0

$$m = (c, a) = 72 + 56 = 128$$

$$k = (u, g) = 56 + 56 = 112$$

A = 1

$$y = (u, c) = 56 + 72 = 128$$

$$r = (a, g) = 56 + 56 = 112$$

A = 2

$$s = (c, g) = 72 + 56 = 128$$

$$w = (u, a) = 56 + 56 = 112$$

$$240 = (128, 112) \rightarrow ((72, 56), (56, 56))$$

$b_1 = Qb_0$ is the phase-bearing sector; $24E_{\text{edge}} = 1$ adds one bonus unit.

Figure 3: Native yellow biological derivation. The 240-site scaffold is derived internally from the quarter alphabet, the triplet selector block, and the normalized edge residual $24E_{\text{edge}} = 1$. The external codon:anticodon / E_8 report is comparison, not input.

5.3 External comparison and scope

The Halitsky–Klitzing–Moxness preprint remains relevant, but only as an external comparison target [21]. The current v2 derivation does not feed its biological counts into the filter. Instead it derives the scaffold internally and then observes that the resulting decomposition agrees with the codon:anticodon / E_8 count pattern reported there.

This sharpened statement matters. The yellow biological observable is a coarse quotient of the lifted state. On the declared biological quotient it is insensitive to $q = (u, v)$ and to the sheet index κ . That is not a contradiction of the earlier negative controls, because those controls targeted finer readouts whose exact covariance depends on ordered-pair and sheet retention. Here the biological observable sits later in the quotient chain and intentionally forgets more data.

What is now closed is stronger than in v1:

- (i) the yellow readout is attached to the same retained lifted state as green, blue, and red;
- (ii) the yellow local section reproduces the CF01 QGT split on its declared chart;
- (iii) the yellow biological section derives the 240-site scaffold and the refinement

$$240 = (128, 112) \rightarrow ((72, 56), (56, 56))$$

without importing those counts as numerical input.

Not yet closed:

- (i) uniqueness of the biological projection among all conceivable yellow projectors,
- (ii) uniqueness of the cyclic RNA naming relative to every other biological labelling convention,
- (iii) a full biochemical dynamics law descending from the same scaffold,
- (iv) a proof that the count-level scaffold already determines the full E_8 root geometry rather than only matching its decomposition surface.

6 Unification Theorem

Theorem 2 (Unification theorem (demonstration form, v2)). *Let*

$$\widehat{\Xi} = (A, q, \theta, \kappa, c), \quad q = (u, v), \quad c = \left[\theta - \frac{\pi}{uv}, \theta + \frac{\pi}{uv} \right],$$

be the exact lifted state on the balanced-window family, and let $\mathcal{E}$ be the single parent evolution generated by the Phase Calculus operator spine and its selector-closed executable macro-calculus. Then, on the declared regimes closed by the merged formalisation, the four-filter package, and the native biological supplement:

- (i) *the green, blue, red, and yellow readouts are covariant projections of the same $\widehat{\Xi}$ under the same operator grammar;*
- (ii) *the yellow quotient readout reproduces the CF01 quantum-geometric-tensor split into $g_{\mu\nu}$ and $\Omega_{\mu\nu}$ on its declared local chart;*
- (iii) *the later physics branches of the current CF stack descend from the same parent evolution by quotient projection and derivation-chain commutation on their declared regimes; and*
- (iv) *the yellow biological readout derives, from the quarter alphabet generated by $\mathbf{Q}$, the triplet selector block, and the normalized edge residual $24E_{\text{edge}} = 1$, the scaffold*

$$240 = (128, 112) \rightarrow ((72, 56), (56, 56)).$$

Therefore the current demonstration supports one precise native-source statement: the same singular lifted object $\widehat{\Xi}$ is the common parent object for the present Phase Calculus hierarchy, for the physics branches descended from it, and for the discrete biological scaffold derived on the yellow quotient.

Proof sketch. The operator-core, selector-closed executable macro-calculus, arithmetic-completion bridge, and exact quotient criterion are supplied by the merged formalisation. The four-filter package then evaluates four distinct projection families on the same retained state variables and shows exact-zero primary covariance errors, machine-precision recovery errors, and failure of stripped-state controls on the readouts that still require full lifted-state retention. The yellow filter is constructed so that its continuous local section is exactly the QGT section used in the CF01 bridge. The present v2 paper then derives the discrete biological scaffold from the yellow quotient itself: the quarter alphabet gives four states, the macro block gives $4^3 = 64$ triplet words, the normalized edge residual contributes one extra discrete channel, and the host labels index the three nontrivial $2 + 2$ partitions of the four-cycle. These steps are attacked in the Lean, SymPy, and Jupyter companions. The theorem is intentionally regime-qualified: it asserts native-source unification for the current closed stack and declared projector family, not uniqueness of every possible biological projector or full E_8 geometry from count data alone. $\square$

The theorem should be read together with its regime statement. The merged formalisation already allows exact or effective descent depending on whether the branch law is algebraic or windowed [1]. The four-filter package then adds the biological yellow layer and tests it on the exact same retained lifted state. The demonstration therefore closes native-source unification at the level appropriate to the present evidence: common parent object, common parent evolution, faithful multi-filter readout, and exact internally derived biological decomposition.

7 Implications & Next Steps

The immediate implication is architectural. If the current stack is right, then the correct way to extend Phase Calculus is not to keep inventing new primitive objects branch by branch. It is to keep the primitive state fixed and ask what further projectors, residual laws, or regime windows are forced or ruled out.

For physics, the next technical work is not another broad manifesto. It is sharper quotient work: more explicit projector formulas, tighter residual bounds, and direct simulation-to-branch interfaces for the declared effective regimes. For biology, the next work is even clearer: prove or disprove uniqueness of the yellow biological projector, extend the scaffold test beyond the current anchor/corridor family, and push the mapping against additional codon-position and biochemical datasets.

The negative controls are especially useful here. They show exactly what can go wrong if one strips too much state. Any later biological or physics extension should be required to survive the same style of impossibility witness.

8 Closure Certificate

8.1 Assumption ledger

This paper uses five assumption classes.

1. **Imported formal closure.** The primitive-source layer, the lifted object, the operator-core, the selector-closed executable macro-calculus, the exact quotient theorem, the continuous shadow theorem, and the derivation-chain exactness theorem are imported from the merged formalisation and the underlying *CF* papers [1, 4, 5].
2. **Imported computational closure.** The four-filter gate values, recovery errors, negative controls, and QGT residuals are imported from the executed four-filter package [20].
3. **Native biological derivation.** The quarter alphabet, the triplet selector block, the normalized residual bonus $24E_{\text{edge}} = 1$, the 240 total, the (56, 72, 56, 56) centered counts, and the three host-indexed (128, 112) bifurcations are derived inside the present paper and attacked in the Lean, SymPy, and Jupyter companions.
4. **External comparison benchmark.** The Halitsky–Klitzing–Moxness preprint is used only as an external comparison target for the codon:anticodon / E_8 count pattern [21].
5. **Demonstration-level synthesis.** The present paper contributes the compressed exposition, the cross-linking of the imported results, and the regime-qualified central unification statement.

8.2 Limits

This paper does *not* claim:

- (i) a re-proof of the entire formal stack,
- (ii) uniqueness of the yellow biological map,
- (iii) a full contour-equivalence theorem for the mock-modular branch,
- (iv) universal biological dynamics derived independently from the present scaffold alone,
- (v) the full E_8 root geometry from count data alone.

It also keeps the same discipline as the Shadow Calculus supplement: completion/shadow data remain downstream of the primitive operator core and may not be promoted to source law by rhetoric alone [4].

8.3 Red-team checklist

A reviewer can now challenge the stack in five targeted places.

1. **Lifted-state sufficiency.** Show that the retained state $(A, q, \theta, \kappa, c)$ is not minimal or not sufficient.
2. **Selector closure.** Find a reachable executable balanced-window case not governed by the state-only selector.
3. **Quotient descent.** Produce a declared branch whose projector does not commute with the parent evolution even on its own stated regime.
4. **Four-filter covariance.** Reproduce the package and show a nonzero primary covariance failure, or demonstrate that one of the negative controls does *not* fail when stripped.
5. **Biological derivation.** Show that the native scaffold does *not* follow from the quarter alphabet, the triplet selector block, and the normalized residual bonus $24E_{\text{edge}} = 1$, or show that the distinguished doubled sector is not canonically fixed by the declared yellow chart.

The point of the present paper is that the targets are now specific enough to attack.

A Benchmark tables

Table 4: Key four-filter quantitative results used in the paper

Quantity	Value imported from the executed four-filter package
Canonical anchor	$(u, v) = (55, 89)$, $uv = 4895$, $\delta_* = \pi/4895$, $\text{Im}(\tau_*) = 1/4895$
Primary covariance gates	Max green, blue, red, yellow error = 0 on short exhaustive, deep deterministic, and random-walk tests
Recovery gate	48 samples; largest reported δ -recovery residual $\approx 1.06 \times 10^{-100}$ (green/blue) and 6.98×10^{-105} (yellow)
Negative control 1	Product-only collapse: $(1, 6)$ and $(2, 3)$ share $uv = 6$ but have different B -images $(6, 7)$ and $(3, 5)$
Negative control 2	Sheet-loss witness: Q^4 visible recurrence is insufficient when κ is removed
QGT bridge	15 corridor rows; maximal closed-form residual $\approx 1.49 \times 10^{-101}$
Biological scaffold	Native identities checked in the v2 supplement: $4^3 = 64$, $24E_{\text{edge}} = 1$, weights $(1, 2, 1, 1)$, total 240, centered counts $(56, 72, 56, 56)$, and host partitions $(128, 112)$

B Minimal filter pseudocode

For orientation only, the four filters can be written schematically as

$$\Pi_{\text{green}}(\widehat{\Xi}) \rightarrow (S^3, \text{stereographic}), \quad \Pi_{\text{blue}}(\widehat{\Xi}) \rightarrow (SU(2), S^2),$$

$\Pi_{\text{red}}(\widehat{\Xi}) \rightarrow \tau$ and shadow-attached branch data, $\Pi_{\text{yellow}}(\widehat{\Xi}) \rightarrow (E(t), Q_{\mu\nu}, g, \Omega, \text{native 240-slot scaffold})$. The demonstration theorem does not require these filters to be numerically identical. It requires them to remain faithful readouts of the same retained lifted state under the same operator grammar.

C Native biological derivation table

Table 5: Native biological scaffold identities derived in the v2 supplement

Identity	Exact value
Quarter alphabet size	4
Triplet selector block size	$4^3 = 64$ words
Normalized edge bonus	$24E_{\text{edge}} = 1$
Native weight budget	$(u, c, a, g) = (1, 2, 1, 1)$
Occurrences per base across the triplet block	48
Total scaffold size	$48(1 + 2 + 1 + 1) = 240$
Base-centered scaffold	$(u, c, a, g) = (56, 72, 56, 56)$
Host-indexed bifurcations	$(m, k) = (128, 112), (y, r) = (128, 112), (s, w) = (128, 112)$

References

- [1] Justin K. Lietz. *Phase Calculus: Complete Formalisation of the Primitive Roll, the Exact Lifted Object, the Lifted-State Operator Core, the Selector-Closed Executable Macro-Calculus, the Universal Lifted-State Calculus, and Shadow Calculus*. Supplied merged manuscript, 2026.
- [2] Justin K. Lietz. *Phase Calculus: Complete Formalism — Lifted-State Operator Calculus of the Primitive Roll*. Draft v0.3, 2026.
- [3] Justin K. Lietz. *Phase Calculus: Connection-Closure Proofs*. Note, 2026.
- [4] Justin K. Lietz. *Shadow Calculus: Supplement to Phase Calculus — Completion, Residual, and Launched-Family Branch Structure*. Draft v0.3, 2026.
- [5] Justin K. Lietz. *CF 19: The Full Lifted Object of the Primitive Roll and Exact Return to Lifted Origin*. 2026.
- [6] Justin K. Lietz. *CF 000: Primitive Distinguishability and Orthogonal Articulation*. 2026.
- [7] Justin K. Lietz. *CF 00: Induced Geometry and Emergent Dynamics in VDM*. 2026.
- [8] Justin K. Lietz. *CF 01: Quantum Geometric Tensor to Metriplectic Brackets in VDM*. 2026.
- [9] Justin K. Lietz. *CF 04: Telegraph–Fisher Causality*. 2026.
- [10] Justin K. Lietz. *CF 06: Information Geometry Foundations*. 2026.
- [11] Justin K. Lietz. *CF 07: Measurement Theory Foundations for VDM*. 2026.
- [12] Justin K. Lietz. *CF 08: Spinor Emergence via Domain-Wall Fermions in VDM*. 2026.
- [13] Justin K. Lietz. *CF 09: Gauge Field Emergence via Berry Connection in VDM*. 2026.
- [14] Justin K. Lietz. *CF 12: Gravity, Black Holes, and Saturation Geometry in VDM*. 2026.
- [15] Justin K. Lietz. *CF 14: Principle of Stationary Action as Temporal Effective Invariant*. 2026.
- [16] Justin K. Lietz. *CF 15: Noether’s Theorem as Invariant Articulation Cost Under Symmetry*. 2026.
- [17] Justin K. Lietz. *CF 16: Electroweak Fiber Splitting, Chiral Weak Coupling, Neutral Gauge Mixing, and Broken Gauge–Scalar Structure in VDM*. 2026.
- [18] Justin K. Lietz. *CF 17: Non-Abelian Saturation, Flux-Tube Formation, Quantitative Confinement, String Breaking, and Deconfinement in VDM*. 2026.
- [19] Justin K. Lietz. *CF 18: Scale Re-Articulation, Effective Action, Hidden-Return Corrections, Running Couplings, Renormalization, and Corrected Symmetry Structure in VDM*. 2026.
- [20] Justin K. Lietz. *phase_four_filter_package_v1*. Executed validation package with note, notebook, summary, and gate outputs, 2026.

- [21] David Halitsky, Richard Klitzing, and Greg Moxness. *The H-bond Acceptor Scaffold for Watson–Crick Codon:Anticodon Pairing and the 4_{21} Polytope Scaffold for E_8* . Preprint supplied with the current package, 2026.